

Mohammad Zaid

GenAI Engineer & Architect · 5+ Years · TCS ♀ Lucknow, India

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PROFESSIONAL SUMMARY

GenAI Engineer & Architect with 5+ years at Tata Consultancy Services, specializing in **production-grade LLM systems, agentic pipelines, and RAG architectures**. Delivered end-to-end GenAI products across Healthcare and BFSI domains from LangChain/LangGraph orchestration and automated code modernization to design to code automation using Claude AI. Hands-on with Python, FastAPI, vector databases, and cloud AI APIs. Experienced team lead (4–6 engineers) who bridges architecture, delivery, and client communication.

EXPERIENCE

Tata Consultancy Services · System Engineer (GenAI Architect & Team Lead)

New Delhi | Apr 2021 – Present

◆ Healthcare - GenAI Architect & Team Lead (Python · LangChain · LangGraph · Claude AI · Agentic AI)

- Architected an AI application that auto-generates business logic from code flow diagrams using LangChain, LLMs, and Python, cutting manual analysis time by ~50% across engineering teams.
- Designed production RAG pipelines with vector stores (FAISS) and LLMs for context-aware document retrieval, improving knowledge extraction accuracy over large enterprise codebases.
- Built a legacy Java code reverse engineering solution using Claude AI + LangChain + RAG + custom AST-based parser to auto-generate structured technical documentation, reducing documentation overhead by ~60% and accelerating modernization.
- Developed an end-to-end Figma to React pipeline: Claude AI for semantic design interpretation, Playwright for DOM traversal and visual validation, and internal MUI component library achieving 80%+ design system compliance and eliminating design engineering handoff cycles.
- Designed and built a multi-agent PR review system on LangGraph that automatically analyses pull requests on GitHub, parsing diffs, identifying logic errors, flagging security anti-patterns, and posting structured inline review comments via the GitHub API.

◆ BFSI - Capital Markets Trade Surveillance (KDB+/q · SDL · SDP)

- Optimised a real-time trade surveillance system monitoring FICC trades for misconduct detection under FCA and MAR regulatory frameworks, improving alert precision across multi-asset classes.
- Maintained SDL/SDP pipelines in KDB+/q for high throughput, low-latency tick data ingestion; authored high-performance time-series queries and led root cause analysis for production data integrity issues.

◆ Aviation - Database Management & Optimisation (SQL · Linux)

- Managed and optimised relational databases for aviation operations; built SQL-based ETL pipelines with indexing and partitioning strategies powering analytics-ready dashboards.

Dweepi Infotech · Intern - Systems & Data (Kernel)

New Delhi | Jul 2019 – Dec 2019

- Worked on Linux kernel internals including device driver development, interrupt handling, and memory management.
- Assisted in code review and testing of kernel patches, collaborating with senior engineers on system-level debugging and documentation.

KEY SKILLS

Generative AI & LLMs: LangChain, LangGraph, LangSmith, LlamaIndex, CrewAI, AutoGen, OpenAI API, Anthropic Claude API, Prompt Engineering, RAG, Agentic Workflows, Fine-tuning, MCP Servers

Vector Databases: FAISS, Chroma, Pinecone, OpenAI Embeddings, text-embedding models

Python & AI Tooling: Python, FastAPI, Hugging Face Transformers, Playwright, AST parsing, API integration & automation

MLOps / LLMOps: LangSmith, Docker, Docker Compose, CI/CD, Git, Kafka, Redis Streams

Cloud AI: Azure AI Services / OpenAI on Azure, AWS Bedrock, Vercel

Data & Databases: PostgreSQL, TimescaleDB, Redis, KDB+/q, SQL, ETL pipelines, time-series analytics

Agile & Leadership: Team Lead (4–6 engineers), Jira, Rally, Sprint Planning, cross-functional deliver

EDUCATION

Dr APJ Abdul Kalam Technical University · B. Tech - Computer Science & Engineering

Lucknow | 2015 – 2019

SSV Inter College · Intermediate - Mathematics, Physics, Chemistry

Ayodhya | 2013 – 2015

SSV Inter College · High School - Science

Ayodhya | 2012 – 2013

CERTIFICATIONS

- LangChain for LLM Application Development
- Building Systems with the ChatGPT API
- Generative AI with Large Language Models - AWS
- Microsoft Certified: Azure AI Fundamentals
- KX Advanced Course (Level 1 / 2 / 3) - KX Systems
- Data Analysis with Python – freeCodeCamp

PERSONAL PROJECTS

FileForge(ForgeTech) - Browser-Native AI File Toolkit | Next.js · TypeScript · ONNX Runtime Web · TensorFlow.js · WebAssembly · Tesseract.js · FFmpeg · Tailwind

🔗 <https://forgetech.vercel.app> | GitHub: github.com/mohd98zaid/FileForge

- Built a browser-native platform with 185+ file tools (image, PDF, audio, video, developer utilities) that processes all files entirely client-side using WebAssembly and ONNX - zero file uploads, zero backend, zero privacy risk.
- Integrated AI-powered background removal via ONNX Runtime Web, OCR across 100+ languages via Tesseract.js, and audio/video conversion via FFmpeg WASM - all running in the browser without any server roundtrip.
- Built for Indian users with full Hindi/English bilingual UI, SSC/UPSC/NEET/Railway exam photo presets, and passport/visa photo tools - serving both domestic and international audiences.
- Deployed on Vercel with auto-generated sitemap, SEO metadata per tool page, and PWA support.

RepairDeskz - AI-Powered Repair Shop Management System | Next.js · FastAPI · PostgreSQL · Redis · WhatsApp Business API · Three.js · GSAP · R3F

🔗 <https://repairdeskz.vercel.app> | GitHub: github.com/mohd98zaid/RepairDesk

- Full-stack PWA for independent repair shops covering the complete ticket lifecycle - drop-off, diagnostics, parts tracking, invoicing, digital signature capture, and customer pickup notification.
- Integrated WhatsApp Business API for automated customer notifications on every ticket status change; integrated an AI assistant for natural-language shop analytics ("Which technician closed most tickets this week?").
- Built real-time inventory management with auto-deduction on ticket completion, low-stock alerts, supplier management, and per-part margin tracking.
- Cinematic 3D landing page built with React Three Fiber (R3F), GSAP scroll-driven animations, and Lenis inertial scrolling - showcasing enterprise-grade frontend engineering.

QuantDSS - Quantitative Trading Decision Support System | FastAPI · Python · Next.js · PostgreSQL/TimescaleDB · Redis · Kafka · Docker

🔗 <https://github.com/mohd98zaid/QuantDSS>

- Self-hosted intraday trading platform for NSE/BSE equities with live market data ingestion via Upstox/Angel One WebSockets, 11-layer signal intelligence pipeline, and a 17-rule risk engine with hard circuit breakers.
- Strategies include EMA Crossover, RSI Mean Reversion, ORB+VWAP, Volume Expansion, and VWAP Reclaim - each signal passes deduplication, meta-strategy blocking, ML/NLP filtering, and liquidity checks before reaching the risk engine.
- Chaos-tested to enterprise standard: distributed locking for concurrent signal safety, PEL recovery workers, session-isolated broker retries, zombie position handling, and auto-recovery from Redis exhaustion and broker failures.
- Supports Paper Trading and Live Execution modes with Kafka + Redis Streams dual-write pipeline, Telegram bot notifications, Prometheus/Grafana observability, and full tick data lake for backtesting.